

Lecture 28/Module 3

Sinusoidal Steady State Analysis

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Today's Topic

- AC Power Analysis
- Instantaneous Power
- Average Power
- Apparent Power & Power Factor

Instantaneous Power

$$v(t) = V_m \cos(\omega t + \theta_v)$$

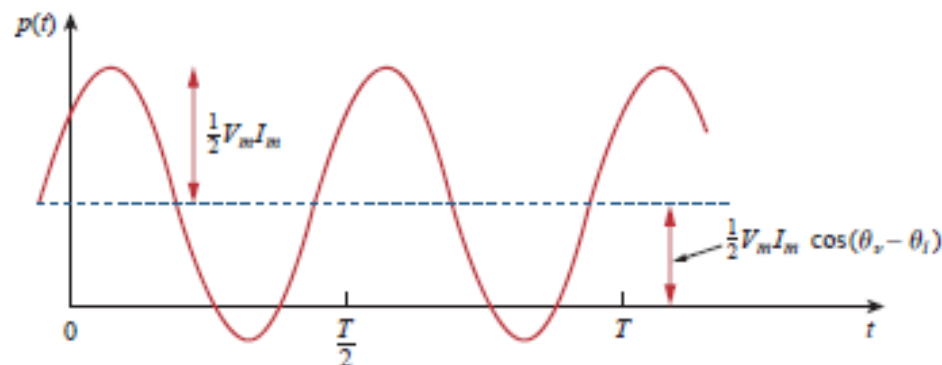
$$i(t) = I_m \cos(\omega t + \theta_i)$$

The instantaneous power (in watts) is the power at any instant of time.

$$p(t) = v(t)i(t)$$



$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$



Average Power

The **average power**, in watts, is the average of the instantaneous power over one period.

$$P = \frac{1}{T} \int_0^T p(t) dt \quad \rightarrow \quad \begin{aligned} P &= \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) dt \\ &\quad + \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i) dt \\ &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \frac{1}{T} \int_0^T dt \\ &\quad + \frac{1}{2} V_m I_m \frac{1}{T} \int_0^T \cos(2\omega t + \theta_v + \theta_i) dt \end{aligned}$$

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

A resistive load (R) absorbs power at all times, while a reactive load (L or C) absorbs zero average power.

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$\begin{aligned} P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) \\ &= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) \end{aligned}$$

$$S = V_{\text{rms}} I_{\text{rms}}$$

The **apparent power** (in VA) is the product of the rms values of voltage and current.

$$\text{pf} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

The **power factor** is the cosine of the phase difference between voltage and current. It is also the cosine of the angle of the load impedance.

$$\mathbf{S} = \frac{1}{2} \mathbf{V} \mathbf{I}^*$$

$$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

$$\mathbf{V}_{\text{rms}} = \frac{\mathbf{V}}{\sqrt{2}} = V_{\text{rms}} \angle \theta_v$$

$$\mathbf{I}_{\text{rms}} = \frac{\mathbf{I}}{\sqrt{2}} = I_{\text{rms}} \angle \theta_i$$

$$\mathbf{S} = I_{\text{rms}}^2 (R + jX) = P + jQ$$

$$\begin{aligned} \mathbf{S} &= V_{\text{rms}} I_{\text{rms}} \angle \theta_v - \theta_i \\ &= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) + j V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i) \end{aligned}$$

1. $Q = 0$ for resistive loads (unity pf).
2. $Q < 0$ for capacitive loads (leading pf).
3. $Q > 0$ for inductive loads (lagging pf).

Thus,

Complex power (in VA) is the product of the rms voltage phasor and the complex conjugate of the rms current phasor. As a complex quantity, its real part is real power P and its imaginary part is reactive power Q .

$$\begin{aligned}\text{Complex Power} = S &= P + jQ = \mathbf{V}_{\text{rms}}(\mathbf{I}_{\text{rms}})^* \\ &= \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}} / \theta_v - \theta_i\end{aligned}$$

$$\text{Apparent Power} = S = |\mathbf{S}| = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}} = \sqrt{P^2 + Q^2}$$

$$\text{Real Power} = P = \text{Re}(\mathbf{S}) = S \cos(\theta_v - \theta_i)$$

$$\text{Reactive Power} = Q = \text{Im}(\mathbf{S}) = S \sin(\theta_v - \theta_i)$$

$$\text{Power Factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

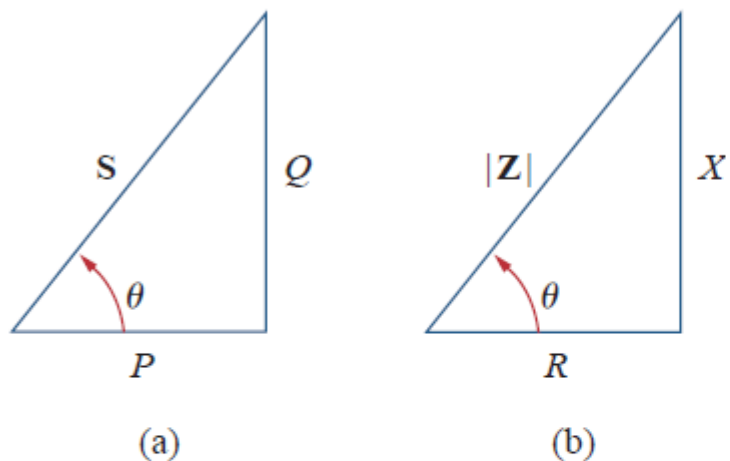


Figure 11.21

(a) Power triangle, (b) impedance triangle.

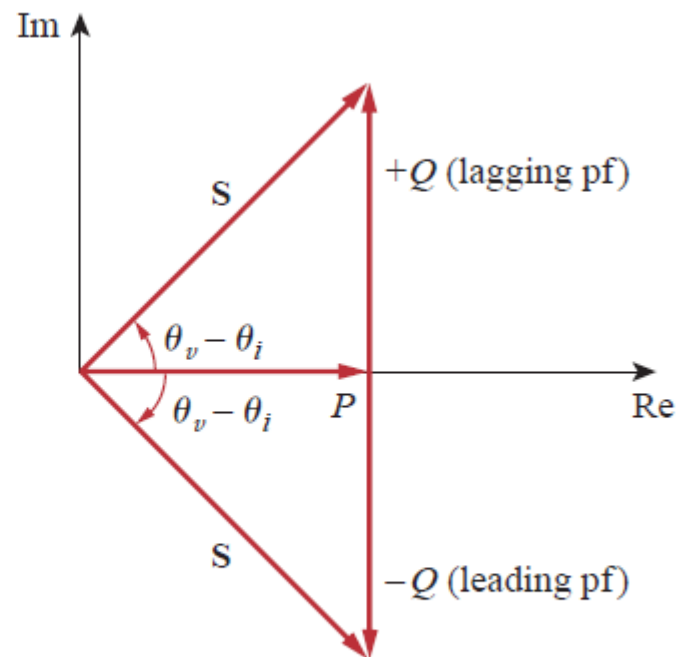


Figure 11.22

Power triangle.

Thank You